

STILLSTACK · CALCULATION

StillStack sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	SSK-DDR-003 accepted (2026-10-02); R4, R7 and R10 rows note the decisions; figures unchanged
0.4	2026-10-01	Amish Chadha	Design for construction (SSK-DDR-003); constructability checks; budget treated as a value-engineering target
0.3	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing (stage energy balance, supports, stagnation, wick feed, salt, mass, tilt, cost)

StillStack sizing calculations

On paper the decided design (four stages, 6 mm gaps, air-cooled fins, 1 m² at 20 deg) makes about **10.6 L per m² per day** on the 5.5 kWh per m² design day, with a gained output ratio of **1.26 to 1.34**. That meets R1 with a thin margin, puts R2 at risk, and misses the 13 L design goal. Dry stagnation reaches about **140 °C** at stage 1 (**144 °C** in calm air). Under SSK-DDR-002 R8 now asks for a 150 °C rating in stages 1 and 2 and 110 °C elsewhere, plus a stagnation cover; the margins are 6 K and 0 K, so R8 is **at risk**. Value-engineering target: USD 355. Estimated cost of the constructable design: USD 416.20 (USD 61.20 over the target). The constructable design of SSK-DDR-003 (base ring, glazing trim, zigzag plate edges with distillate and brine tongues, stop blocks, pinned props) leaves the yield at 10.6 L and the panel at 18.5 kg dry. Every number below is printed by `docs/04-calcs/sizing.py`; the geometry figures come from `cad/src/model.py`.

■ 1. Method

The script solves a quasi-steady energy balance of the stack every 15 minutes through a sinusoidal design day, then integrates.

- **Nodes.** Absorber plate, three intermediate condenser plates and the bottom plate (five plate temperatures), plus the four wet wick faces. Heat conducts from each plate through its 1 mm wick ($k = 0.45 \text{ W/mK}$) to the wick face.
- **Vapor gap.** Latent transfer by Stefan diffusion of vapor through air, $m = (P M D / R T d) \ln((P - p_c)/(P - p_w))$; conduction through still air (the gap is heated from above and stably stratified, so $Nu = 1$); gray-body radiation between the wet wick (emissivity 0.95) and the condensing face (0.90); conduction through the ribs and rails.
- **Top loss.** Absorber to glazing across the 25 mm air layer (Hollands correlation for an inclined layer, plus radiation with emissivities 0.90), across the twin-wall cells (9.3 W/m²K, from a typical sheet U-value of 3.6 W/m²K less surface films), and from the outer sheet to wind ($h = \max(5, 8.6 V^{0.6} / L^{0.4})$) and sky (Swinbank sky temperature).
- **Bottom.** Natural convection from a heated surface facing down at 70 deg from vertical (Fujii and Imura), combined with sheltered wind at half the free-stream speed; fin efficiency for 0.5 mm aluminum fins 30 mm deep; radiation from the bare aluminum underside (emissivity 0.10) to ground at ambient.
- **Feed.** Each stage heats its own feed from ambient to the wick temperature at a feed-to-distillate ratio of 2.5, and that heat leaves with the brine.
- **Edges.** 12 mm plywood plus 25 mm foil-faced stone wool (k about 0.035 W/mK), $U = 1.07 \text{ W/m}^2\text{K}$, over the wall height of each stage. Stone wool replaced PIR under SSK-DDR-002 with the same assumed conductivity, so the energy balance is unchanged.
- **Thermal mass.** Treated separately as a warm-up deduction, because the quasi-steady solution ignores it.

Assumptions

- Design day: 5.5 kWh per m² (19.8 MJ per m²) on the tilted aperture, spread as a sine over 10 h (peak 864 W/m²); ambient 30 °C; wind 1 m/s.
- Optics: twin-wall polycarbonate transmittance 0.80 and absorber absorptance 0.95, so $\tau\alpha = 0.76$. The TRL 2 estimate used 0.88 for the glazing, which is a single-glass value.
- Latent heat varies with temperature (2.33 to 2.36 MJ per kg over the stage range), replacing the flat 2.26 MJ per kg used at TRL 2.

- Wet wick area: wicks are cut into five strips with 10 mm dry breaks at each rib and rail (R4) and stop 20 mm short of the low edge, leaving 82.4 % of the aperture wet.
- The feed ratio is held at the R6 minimum of 2.5 to limit sensible loss.
- Properties of air and vapor use standard power-law fits; saturation pressure uses the Buck equation.

■ 2. Vapor gap transfer

Most of the heat crossing each gap is latent, and the fraction rises steeply with temperature (Table 1). The TRL 2 assumption of 0.70 per stage was conservative: at the noon operating point the four stages run at 0.91, 0.89, 0.87 and 0.85.

Table 1. Heat flux across one 6 mm gap with 5 K between the wick face and the condensing face.

Wick face (°C)	Latent (W/m ²)	Conduction (W/m ²)	Radiation (W/m ²)	Evaporation fraction
40	150	22.3	29.2	0.74
50	251	22.9	32.1	0.82
60	418	23.4	35.2	0.88
70	712	24.0	38.5	0.92
80	1,298	24.6	42.0	0.95

■ 3. Plate support and spacer ribs

A 0.5 mm aluminum plate carrying a wet wick cannot span the frame. Under 22.6 N/m² (plate, 1.0 kg/m² of wet wick and condensate, normal to the 20 deg slope), beam theory gives a sag of about 325 mm over the 0.972 m clear span, so the TRL 2 concept with side rails only would close every gap. Treating each strip as simply supported, a sag of 1.0 mm (one sixth of the gap) allows a pitch of up to 229 mm. **Four intermediate ribs per gap at 194 mm pitch** give a sag of 0.52 mm. The ribs run down the slope so they do not block feed or condensate. Each plate grows about 1.4 mm across 1 m at a 60 K rise, so the model leaves 2 mm of clearance each side instead of tensioning the plates.

Support of the whole stack (SSK-DDR-003). The stack rests on a 12 mm plywood base ring whose 15 mm ledge carries the bottom plate's edges and the side rails. The ribs carry their share of the stack (about 108 N/m² normal to the plate at 20 deg) onto the bottom plate, which spans the 970 mm between the high and low ledges as ten fin-stiffened strips: with the fins bonded the sag is about 0.5 mm, and 1.0 mm if the fins carried the load alone (section 13). The whole stack moves together, so the gaps do not change. The wicks now stop 13 mm inside the plates' zigzag low edge, which lowers the wet wick fraction from 0.824 to 0.817; the design-day yield stays at 10.6 L.

The ribs are 7 mm food-grade silicone cord (decided by Amish on 2026-09-25, SSK-DDR-002); silicone is rated well above the stagnation temperature and wets poorly, which limits brine creep across the rib. Printed polycarbonate ribs would cost about the same.

■ 4. Design day

Table 2. Daily energy and water balance per m² of aperture.

Quantity	Value
Solar input	19.8 MJ
Optical loss	4.8 MJ
Absorbed	15.1 MJ
Top loss	4.5 MJ
Edge loss	0.15 MJ
Heat into stage 1, 2, 3, 4 wicks	10.5, 9.2, 8.1, 7.3 MJ
Feed heating lost with brine, stages 1 to 4	1.32, 1.03, 0.81, 0.63 MJ (3.8 MJ total)
Heat rejected at the bottom	6.6 MJ (peak 277 W/m ²)
Distillate, stages 1 to 4	3.44, 2.97, 2.59, 2.28 L
Distillate, quasi-steady	11.3 L
Gained output ratio, quasi-steady	1.34
Warm-up energy (25.7 kJ/K per m ² of stack to the mean noon temperature)	0.92 MJ
Distillate after warm-up deduction	10.6 L
Gained output ratio after warm-up deduction	1.26

At noon the plates run at 73.3, 69.5, 65.8, 62.2 and 58.4 °C, the bottom plate 28.4 K above ambient, and the top loss is 172 W/m². The TRL 2 figures of 13.2 L and a gained output ratio of 1.5 were optimistic mainly because of the glazing transmittance and because the sensible heat of the feed (3.8 MJ, about 19 % of the sun) was not counted. The warm-up deduction is conservative, since part of the stored heat still produces distillate as the stack cools in the evening.

Table 3. Sensitivity of daily output (quasi-steady).

Case	Distillate (L/m ² /day)	Gained output ratio	Peak absorber (°C)	Peak bottom (°C)
3 stages	9.1	1.08	73	62
4 stages (base)	11.3	1.34	73	58
6 stages	14.9	1.77	74	53
No rib breaks in the wicks	11.7	1.39	72	59
Feed ratio 3.5	10.5	1.25	71	55
Wind 3 m/s	11.6	1.38	69	48
Painted underside (emissivity 0.90)	11.7	1.39	70	50
Sunny day, 6.5 kWh/m ²	13.6	1.36	77	62
Hazy day, 4.5 kWh/m ²	9.0	1.31	69	54

Air fins are adequate for four stages: the bottom plate stays near 58 °C and the stage temperatures stay well below the limits of the wicks and spacers in wet operation. Six stages would gain about 3.6 L on paper without a water-cooled bottom, but the decision is four.

■ 5. Hot operation and dry stagnation

At 1,000 W/m² and 35 °C ambient, wet operation peaks at 80.1 °C at the absorber and 67.1 °C at the bottom plate. With dry wicks the plates reach **140, 123, 106, 87 and 65 °C** (absorber first). In calm air they reach **144, 128, 110, 91 and 69 °C**. The inner glazing sheet reaches about 109 °C (115 °C in calm air) and the outer sheet 61 °C.

Safety: A dry panel in full sun holds the absorber near 140 °C. Fit the stagnation cover before opening the stack, and treat an empty feed container as a burn and material-damage hazard.

Consequences for R8 as restated in SSK-DDR-002. Stages 1 and 2 (absorber to plate 2) peak at 144 °C against the 150 °C rating, a margin of 6 K; their rails are therefore a 150 °C polymer (PPS or a high-heat polycarbonate copolymer), the ribs silicone and the liner stone wool, and the wick and the coating on plates 1 and 2 must tolerate 150 °C. Stages 3 and 4 peak at 110 °C in calm air against the 110 °C rating, a margin of 0 K; printed polycarbonate rails remain acceptable there, but ASA (softening near 95 to 100 °C) stays excluded from stages 1 to 3. The inner glazing sheet at 115 °C in calm air is close to the typical limit of twin-wall polycarbonate. The stagnation cover (BOM line 14) and the shade rule are the second barrier: the cover is fitted whenever the feed runs out.

■ 6. Wick feed capacity

At noon stage 1 evaporates 0.58 kg/m²/h, so each metre of wick width must carry 1.44 kg/h of feed at a ratio of 2.5. With gravity along the 20 deg slope (3,360 Pa/m) and an assumed 4 kPa of capillary pressure over the 1 m run, Darcy flow through a 1 mm wick needs a permeability of at least **3.0 x 10⁻¹¹ m²**. Capacity is 0.48, 1.44 and 4.81 kg/h per metre at 1 x 10⁻¹¹, 3 x 10⁻¹¹ and 1 x 10⁻¹⁰ m². The permeability of the candidate fabrics is not known, so feed supply is a risk to R1 and R6.

■ 7. Salt balance and carryover

At a feed ratio of 2.5 the brine leaves at 58.3 g/L for 35 g/L seawater and 66.7 g/L for 40 g/L feed, a factor of 4.8 below sodium chloride saturation (about 317 g/L). Salt does not travel with the vapor, so distillate salinity depends only on liquid carryover. To stay below 50 mg/L the brine reaching the distillate must be under 0.075 % of the distillate, about **8.5 mL per day** on the design day. On the design day the panel takes about 28.2 L of feed and discharges 16.9 L of brine.

■ 8. Mass, footprint and tilt

Table 4. Panel mass (constructable design, SSK-DDR-003).

Item	Mass (kg)
Plates, five 0.5 mm aluminum, with tongues	6.94
Glazing, twin-wall polycarbonate, 1.06 m square	1.46
Walls, 12 mm plywood, less openings	1.35
Liner, 25 mm stone wool	0.69
Base ring, 12 mm plywood	1.40
Glazing trim, aluminum angle	1.00
Wicks, dry, with tails	1.12
Side rails, 150 °C polymer and printed polycarbonate	0.81
Ribs, silicone cord	0.71
Fins	0.63
Trough, manifold, gutter, brackets, closure and fittings	1.80
Sealant, tape, fasteners and tubing	0.60
Panel, dry	18.5
Panel, wet wicks	22.5
Adjustable stand	8.9

The model's own volume-based estimate for the modeled parts (16.1 kg without the trough, manifold, gutter and fasteners) agrees within the allowances above.

The stand (SSK-DDR-003) pivots the panel on two M10 bolts through front posts 372 mm above the ground and holds it with two fixed props, 1,000 mm between pins, hinged on blocks near the high end and pinned to the ground rails. One hole per tilt sets the angle; measured back from the pivot the holes are at 113, 151, 195, 245, 309 and 396 mm for 10, 15, 20, 25, 30 and 35 deg. The plan footprint is 1.28 x 1.25 m at 10 deg, 1.21 x 1.25 m at 20 deg and 1.05 x 1.25 m at 35 deg. The ground rails run 760 mm behind the pivot, so the wet panel's centre of mass (about 500 mm behind the pivot at 10 deg) stays at least 260 mm inside the ground frame. The model builds and checks the stand at all six holes; the feed trough rim is at most 1.00 m above the ground.

■ 9. Wind

At 20 m/s the normal force on the 1.15 m² panel (force coefficient 1.2) is about 332 N, against 308 N for the wet panel and stand. Lift equals weight at about 18.7 m/s.

Safety: The panel and stand can overturn in a strong breeze. Anchor the stand to the ground or ballast it before filling.

■ 10. Cost

Value-engineering target: USD 355 (a hypothetical control target, `budget_usd` in `project.yaml`). Estimated cost of the constructable design: USD 416.20 from `bom/bom.csv` (18 lines), USD 61.20 over the target. Before SSK-DDR-003 the concept priced at USD 351.20. The USD 65 added for construction is USD 20 for larger plate blanks for the tongues (lines 3 and 5), USD 2 for longer wicks (line 4), USD 20 for the glazing trim and tape (line 15), USD 7 for the stand's ground frame, gussets and pins (line 12), USD 4 for the trough brackets (line 8), USD 6 for the outlet brackets (line 16) and USD 6 for the feed closure (line 17).

■ 11. Results against requirements

Table 5. Requirement status at TRL 3, not met first.

ID	Quantity	Value	Target	Status
R8	Temperatures at dry stagnation	Stages 1 and 2: 144 °C calm (6 K margin); stages 3 and 4: 110 °C calm (0 K margin); cover specified	150 °C in stages 1 and 2, 110 °C elsewhere, cover and shade rule	At risk
R2	Gained output ratio	1.34 quasi-steady; 1.26 after warm-up deduction	1.3 or more	At risk
R4	Air break between wick edges and distillate paths	10 mm at ribs and rails and inside the plate edges; brine tongues pass 10.5 mm or more over the closed manifold lid and at least 44 mm from any distillate tongue (SSK-DDR-003 A1, accepted 2026-10-02; high-edge lip of A2 (c) decided, not yet modelled)	10 mm or more, no shared drain	At risk
R6	Brine salinity; wick feed	66.7 g/L at 40 g/L feed; needs permeability of 3.0×10^{-11} m ² or more	Feed ratio 2.5 or more, salt stays in solution	At risk
R7	Food-contact wetted surfaces	Coated aluminum decided; coating product not selected (rule decided 2026-10-02: certified for drinking-water contact and 150 °C dry, else 316 stainless for plates 1 and 2)	All distillate-side surfaces food-contact rated	At risk
R10	Wick removal time; flushing interval	Plates lift out after the low wall cap is removed; wicks to be held by end clips with no adhesive (decided 2026-10-02)	30 min or less; weekly or less often	At risk
R3	Distillate conductivity	Carryover allowance 0.075 % (8.5 mL/day)	75 µS/cm or less	Not verifiable at TRL 3
R1	Daily distillate	10.6 L/m ² (11.3 quasi-steady)	10 L/m ² or more (goal 13)	Met (6 % margin; goal not met)
R12	Parts cost	USD 416.20	Value-engineering target USD 355	USD 61.20 over the target
R5	Power; feed head	Passive; feed trough rim 1.00 m at 35 deg	No power; container at most 1.5 m	Met
R9	Mass; footprint	18.5 kg dry; 1.28 x 1.25 m	20 kg or less; within 1.3 x 1.3 m	Met
R11	Tilt range	Six prop holes, 10 to 35 deg in 5 deg steps; model built and checked at each	10 to 35 deg	Met

■ 12. Changes to earlier figures

SSK-PRC-001 v0.2 gave 13.2 L per day, a gained output ratio of about 1.5, stage 1 at 70 to 80 °C, dry stagnation at 100 to 120 °C, feed of 35 L and brine of 22 L, a panel of about 15 kg and parts of about \$241. The precis v0.3 now quotes the values in this note.

Version 0.2 of this note (SSK-DDR-002) adds the calm-air plate temperatures and the restated R8 check, and changes the panel mass from 15.9 to 16.7 kg dry (19.9 to 20.7 kg wet), the wet panel and stand weight from 264 to 272 N, and the parts cost from \$317.20 to \$351.20 against a budget raised from \$250 to \$320. The energy balance, yield and gained output ratio are unchanged.

Version 0.3 of this note records the budget top-up to \$355 approved by Amish on 2026-09-26; the parts cost is unchanged at \$351.20 and R12 moves from not met to met.

Version 0.4 of this note follows SSK-DDR-003 (design for construction, made under Amish's 2026-09-30 instruction to make the design physically buildable; accepted by Amish on 2026-10-02 with the exceptions recorded there). The wet wick fraction falls from 0.824 to 0.817, which moves the stage outputs by at most 0.01 L; the daily yield (10.6 L), gained output ratio and stagnation temperatures are unchanged at the precision quoted. The panel rises from 16.7 to 18.5 kg dry (20.7 to 22.5 kg wet), the stand from 7.1 to 8.9 kg, the footprint from 1.28 x 1.16 m to 1.28 x 1.25 m, the wind speed at which lift equals weight from 17.5 to 18.7 m/s, and the cost from USD 351.20 to USD 416.20, now reported against the USD 355 value-engineering target. Feed and brine figures were also brought in line with the script (28.2 and 16.9 L per day).

■ 13. Design for construction checks

Table 6. Checks added for the constructable design (section 14 of the script).

Check	Result
Bottom plate sag on the base ring, fins bonded (composite)	0.5 mm
Bottom plate sag, fins carrying the load alone	1.0 mm
Glazing growth, 1.06 m sheet at a 70 K rise	4.8 mm, against 14 mm of room inside the trim
Load per pivot bolt, wet panel plus 17 m/s wind	163 N; bearing on 12 mm plywood 1.4 MPa
Prop buckling, 45 x 45 mm timber, 1.0 m pinned	27 kN, far above the load
Outlet bracket load, manifold and gutter full	29 N each; 62 N pull on the top screw
Vapor lost through the low-end tongue openings (diffusion, 4 gaps)	25 mL per day (0.2 % of the yield)
Lowest brine tongue tail above the distillate lid	10.5 mm

The parametric model also runs its own constructability checks (`python cad/src/model.py --check`): no overlaps among the 46 panel parts, every part that must touch does touch, every brine path keeps 10 mm or more from every distillate path, and the stand at each of the six tilt holes has no clashes, keeps the centre of mass inside the ground frame and stays within the 1.3 x 1.3 m footprint.